



ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

Model Syllabus for Quantum Technologies (Minor) in consonance with Curriculum framework w.e.f. AY 2025-26

COURSE STRUCTURE

Year	Semester	Course	Title of the Course	No. of Hrs /Week	No. of Credits
II	III	1	Mathematical Physical and Computing Foundations of Quantum Computing	5	4
	IV	2	Quantum Technologies: Principles and Applications	5	4
III	V	3	Foundations of Quantum Technologies	5	4
		4	Numerical Methods for Quantum Applications	5	4
	VI	5	Quantum Algorithms	5	4
		6	IBM Quantum Composer and Quantum Computing using Qiskit Lab	5	4

SEMESTER-III

COURSE 1: MATHEMATICAL PHYSICAL AND COMPUTING FOUNDATIONS OF QUANTUM COMPUTING

Theory

Credits: 4

5 hrs/week

Course Outcomes

At the end of the course, students will be able to

CO1: Students will be able to compute eigenvalues, eigenvectors, and perform diagonalization of matrices, and interpret these results in the context of quantum state representation.

CO2: Students will be able to model random experiments using probability distributions and apply the Central Limit Theorem to analyze data relevant to quantum mechanical systems.

CO3: Students will be able to apply Maxwell's equations to derive wave propagation characteristics in free space, conducting media, and optical fibers.

CO4: Students will be able to trace the instruction execution cycle in a microprocessor, including fetch, decode, and execute phases, and explain the role of memory hierarchy in system performance.

CO5: Students will be able to design optimized combinational circuits such as adders, multiplexers, and code converters using Boolean algebra and Karnaugh map simplification.

Unit 1. Linear Algebra

Vectors and Vector Spaces, Linear Transformations, Complex Vectors and Matrices, Eigenvalues and Eigenvectors, Inner Product Spaces, Orthogonality and Hilbert Spaces, Diagonalization, Applications to Quantum computing

Unit 2. Probability , Statistics and basics of Quantum Mechanics

Introduction to Statistics, Data Representation, Descriptive Statistics, Probability, Random Variables and Probability Distributions, Specific Probability Distributions and the Central Limit Theorem

Classical Mechanics Overview, Lagrangian Formulation, Hamiltonian Mechanics, Applications to Quantum Mechanics, Introduction to Electromagnetic theory, MAXwell's equations.

Unit 3. Computer Architecture

Digital Number Representation : Number systems: binary, octal, decimal, hexadecimal., Conversion between number systems, Signed number representation: sign-magnitude, 1's complement, 2's complement, Floating-point representation (IEEE 754 basics)., Binary arithmetic (addition, subtraction, multiplication, division).

Computer Architecture Basics: Fundamentals of computer design, Functional units of a computer, Data path and control path, Input-Output organization

Instruction cycle: Fetch, Decode, Execute, Control unit.

Unit 4. Microprocessors and Memory

Microprocessor Fundamentals :Evolution of microprocessors (8085, 8086, Pentium, ARM), CPU internal architecture (ALU, registers, buses)., Addressing modes and instruction formats, Instruction execution and timing, Examples of simple assembly programs.

Memory Hierarchy: Need for memory hierarchy, Cache memory: mapping techniques and replacement policies, Main memory (RAM, ROM, DRAM, SRAM)., Virtual memory concepts (paging, segmentation), Secondary storage overview.

Unit 5. Digital Logic, Boolean Algebra, and Combinational Circuits

Introduction to Digital Logic Gates : Basic gates: AND, OR, NOT, Universal gates: NAND, NOR, XOR, XNOR functions, Truth tables and circuit symbols, Logic gate implementations using transistors.

Boolean Algebra and Simplification : Boolean postulates and theorems, Canonical forms (SOP, POS), Simplification techniques: Boolean identities, algebraic manipulation, Karnaugh maps (2, 3, 4 variables).

Combinational Circuits: Half adder and full adder, Subtractors, comparators, Multiplexers, demultiplexers, Encoders, decoders, Design examples: Arithmetic circuits, code converters.

Textbooks:

1. Bernard Kolman, David A Hill, Elementary Linear Algebra with Applications, Pearson New International Edition, 2013.
2. Ron Larson, Elementary Statistics: Picturing the World, 8th Edition, Pearson, 2023.
3. Herbert Goldstein, Classical Mechanics, 3rd Edition, Addisen Wesley Publisher, 2001
4. Griffiths D. J., Introduction to Electrodynamics, 4th Edition, Cambridge University Press, 2020.
5. Thomas L. Floyd, Digital Fundamentals, 11th Edition, Pearson Publication, 2014.

Reference Books:

1. K. Hoffman and R. Kunze, Linear Algebra, Pearson Education, 2nd Edition, 2005.
2. Erwin Kreyszig, Advanced Engineering Mathematics, Wiley India, 10th Edition, 2015.
3. David Morin, Introduction to Classical Mechanics, 1st Edition, Cambridge University, 2008.
4. William H. Hayt and John A. Buck, Engineering Electromagnetics, 8th Edition, McGraw-Hill Education, 2011.
5. David A. Patterson and John L. Hennessy, Computer Organization and Design: The Hardware/Software Interface, 5th Edition, Morgan Kaufmann, 2013.

SEMESTER-IV

COURSE 2: QUANTUM TECHNOLOGIES: PRINCIPLES AND APPLICATIONS

Theory

Credits: 4

5 hrs/week

Course Outcomes

At the end of the course, students will be able to

CO1: Understand foundational principles governing quantum behavior and contrast them with classical systems, study and interpret evolution and measurement of simple systems.

CO2: Analyze various physical platforms proposed or used for realizing advanced computing technologies.

CO3: Understand the principles behind high-precision measurement techniques enabled by quantum effects.

CO4: Apply coherence and entanglement concepts to real-world sensing applications in navigation and metrology.

CO5: Analyze communication systems using non-classical principles to enhance security and information transfer.

Unit 1: Introduction to Quantum Physics and Quantum Technologies

Motivation for Quantum Technologies, Quantum States, Wave functions, Probabilistic interpretation, Physical observables, Hermitian operators, Expectation values, Heisenberg uncertainty principle, Schrodinger equation, Time evolution, distinction from classical physics, Heuristic description of Superposition, Tunnelling and entanglement, No cloning theorem, Simulating classical systems – Feynman's idea of a quantum simulator and the birth of the field.

Unit 2: Quantum Computation

Basics of Qubits, Superpositions, Entanglement, Measurement and Errors. Difference between a classical bit and quantum Qubits, Review of classical logic gates, Di Vincenzo criteria for realising Qubits, Basics of Qubit gates and quantum circuits Overview of applications and recent achievements: RSA and Shor's algorithm and Quantum Advantage.

Unit 3: Quantum Sensing

Basics of quantum sensing: Definition and scope, Classical vs. quantum sensing and Applications in precision measurements, navigation, and defence, Role of quantum coherence and entanglement in sensing; Principles of single-photon generation, Sources of single photons, Generation of entangled photon pairs. Quantum materials and their properties.

Unit 4: Quantum Communications

Basics of digital communication: Introduction to classical communication systems, Analog vs. digital communication, Encoding, modulation and transmission; Quantifying classical information: Definition and significance of entropy in communication and Shannon entropy; Quantum communication: Principles of quantum information transfer, Qubits in communication systems, Quantum superposition and entanglement in communication, Role of quantum states in secure information transfer;

Unit 5. Simulators and Security

Running quantum protocols in a quantum simulator, Implementing simple quantum algorithms on cloud-based quantum computers.

Security: Quantum Security and Eavesdropping: No-cloning theorem and secure communication, Detection of eavesdropping using quantum principles.

Textbooks

1. R. Shankar, Principles of Quantum Mechanics, 2nd Edition, Springer, 2012.
2. Nouredine Zettili, Quantum Mechanics: Concepts and Applications, 2nd Edition, Wiley, 2009.
3. Michael A. Nielsen and Isaac L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary Edition, Cambridge University Press, 2010.
4. David J. Griffiths and Darrell F. Schroeter, Introduction to Quantum Mechanics, 3rd Edition, Cambridge University Press, 2018.
5. Robert Eisberg and Robert Resnick, Quantum Physics of Atoms, Molecules, Solids, Nuclei and Particles, 2nd Edition, Wiley, 1985.

Reference Books

1. J.J. Sakurai and Jim Napolitano, Modern Quantum Mechanics, 2nd Edition, Cambridge University Press, 2017.
2. Marlan O. Scully and M. Suhail Zubairy, Quantum Optics, 1st Edition, Cambridge University Press, 1997.
3. P. Hemmer, Quantum Technologies and Applications, 1st Edition, Springer, 2020.
4. Eleanor G. Rieffel and Wolfgang H. Polak, 1st Edition, MIT Press, 2011.
5. Mark M. Wilde, Quantum Information Theory, 2nd Edition, Cambridge University Press, 2017.

SEMESTER-V

COURSE 3: FOUNDATIONS OF QUANTUM TECHNOLOGIES

Theory

Credits: 4

5 hrs/week

Course Outcomes

At the end of the course, students will be able to

CO1: Understand the fundamental postulates of quantum mechanics and key principles

CO2: Apply the Schrödinger equation and associated formalism to basic quantum systems

CO3: Analyze quantum statistical methods using Gibbs principle and derive statistical distributions

CO4: Understand classical and quantum information concepts including Shannon entropy, digital encoding, decoherence, and quantum noise.

CO5: Apply basic quantum information theory using Kraus operators and evaluate simple quantum error correction scenarios and study quantum complexity

Unit 1. Quantum Mechanics

Hamiltonian function and Hamilton's equations, Phase-space description of a system, Connection and Equivalence with Newton's laws for simple systems – free particle, particle moving in a conservative potential, examples of Harmonic oscillator, hydrogen atom, Historical evolution of quantum mechanics, Planck's quantum hypothesis, Photo electric effect, Atomic spectra, Bohr's quantisation principle, De Broglie's Wave particle duality, Postulates of Quantum Mechanics, State vectors and Hilbert Space, Dirac Bra-Ket notation, Measurables and Hermitian Operators, Unitary Transformations, Schrodinger Equation and Time evolution of quantum states, Measurement Postulate, Schrodinger, Heisenberg and Interaction pictures, Eigen values, Expectation values and Matrix elements, Heisenberg's Uncertainty principle

Unit 2: Advanced Quantum Mechanics

Density operator formalism of quantum mechanics – pure and mixed states, Superposition and Entanglement in quantum mechanics, No cloning theorem, Applications of postulates –Particle in a box, Hydrogen atom, Harmonic Oscillator, Number states, ladder operators and Coherent states of a harmonic oscillator, Spin and Angular momentum – spin half particles, Rabi problem of a spin-half particle in a rotating magnetic field, Bosons and Fermions

Unit 3. Statistical Physics

Quick review of first and second laws of thermodynamics, Thermal Equilibrium and Gibbs principles, Applying Gibbs principle to Classical and Quantum harmonic oscillators, Bosons and Fermions and Quantum statistics – Fermi-Dirac and Bose-Einstein distributions

Unit 4. Information Science

Digital communication and information, Quantifying information in terms of Shannon entropy, Basic ideas of quantum information, Decoherence and noise, Introductory ideas of Kraus operators

Unit 5. Brief overview of Computational Complexity

Qualitative ideas of a Turing machine, Types of Turing machines, Time and Space complexity – P vs NP, PSPACE, Quantum complexity classes – Q, EQP, BQP, BPP, QMA, Post Quantum Cryptography (PQC)

Textbooks

1. David J. Griffiths, Darrell F. Schroeter, Introduction to Quantum Mechanics, 3rd Edition, 2018.
2. R. K. Pathria, Paul D. Beale, Statistical Mechanics, 3rd Edition, 2011.
3. Michael A. Nielsen, Isaac L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary Edition, 2010.
4. Thomas M. Cover & Joy A. Thomas, Elements of Information Theory, 2nd Edition, 2006.
5. Michael Sipser, Introduction to the Theory of Computation, 3rd Edition, 2012.

Reference Books

1. J. J. Sakurai, Jim Napolitano, Modern Quantum Mechanics, 2nd Edition, 2017.
2. Kerson Huang, Statistical Mechanics, 2nd Edition, 1987.
3. Mark M. Wilde, Quantum Information Theory, 2nd Edition, 2017.
4. Sanjeev Arora, Boaz Barak, Computational Complexity: A Modern Approach, 1st Edition, 2009.
5. Scott Aaronson, Quantum Computing Since Democritus, 1st Edition, 2013

SEMESTER-V

COURSE 4: NUMERICAL METHODS FOR QUANTUM APPLICATIONS

Theory

Credits: 4

5 hrs/week

Course Outcomes

At the end of the course the student will be able to

CO1. Develop efficient programs using data structures, object-oriented concepts, and simple algorithms

CO2. Apply numerical methods to solve ordinary and partial differential equations,

CO3. Implement numerical techniques for matrix inversion, eigenvalue problems, least squares fitting, error analysis, and Monte Carlo sampling.

CO4. Simulate and analyze quantum mechanical systems

CO5. Model and simulate electromagnetic problems

Unit 1. Basics of programming and simple algorithms

Data structures, classes, Object-oriented programming, Data storage and retrieval, Memory allocation, Scientific plotting, documentation of codes

Simple algorithms and benchmarking run time, Sorting, Searching, Arithmetic algorithms like GCD, Prime factorisation

Unit 2. Numerical Integration and differential equations

Linear 2nd Order ODEs with constant coefficients, Linear 2nd order ODEs with variable coefficients, Boundary value problems, Poisson equation, Laplace equation, Wave equation, Diffusion Equation

Unit 3. Numerical techniques in linear algebra, probability and statistics

Numerical techniques in linear algebra: Matrix inverse, Eigenvalue problem, Diagonalisation of matrices, Singular value decomposition

Numerical techniques in Probability and Statistics: Random number generation, Computing statistical moments for data samples, Least Squares fitting, Error Analysis, Hypothesis Testing, Monte Carlo sampling

Unit 4. Applications to Quantum Mechanics (can be done using openly available modules in languages like Python, Julia etc.)

Eigen energies of coupled two level systems, Eigen energies of two-level system coupled to oscillator (Jaynes-Cummings Model), Driven two-level system: Rabi Problem, Driven damped oscillator, coherent states

Unit 5. Applications to EM theory (e.g. magnetic field simulation)

Electrostatic charge distributions, Magnetostatic current distributions, Finite Element techniques for electromagnetic simulations

Text Books

1. Nicholas Giordano, Hisao Nakanishi, Computational Physics, 2nd Edition, Pearson-Addison Wesley, 2005
2. E. Balagurusamy, Object-Oriented Programming with C++, 7th Edition, McGraw Hill Education, 2017
3. Steven C. Chapra, Raymond P. Canale, Numerical Methods for Engineers, 8th Edition, McGraw Hill Education, 2020
4. Gilbert Strang, Linear Algebra and Its Applications, 5th Edition, Cengage Learning, 2016
5. Mark Newman, Computational Physics, Revised and Expanded Edition, Cambridge University Press, 2013
6. David J. Griffiths, Darrell F. Schroeter, Introduction to Quantum Mechanics, 3rd Edition, Cambridge University Press, 2018

Reference Books

1. William H. Press, Saul A. Teukolsky, William T. Vetterling, Brian P. Flannery, Numerical Recipes: The Art of Scientific Computing, 3rd Edition, Cambridge University Press, 2007.
2. John David Jackson, Classical Electrodynamics, 3rd Edition, Wiley, 1998.
3. Michael A. Nielsen, Isaac L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary Edition, Cambridge University Press, 2010.
4. Arieh Iserles, A First Course in the Numerical Analysis of Differential Equations, 2nd Edition, Cambridge University Press, 2008.
5. Hans Petter Langtangen, Svein Linge, Programming for Computations – Python: A Gentle Introduction to Numerical Simulations with Python, 2nd Edition, Springer, 2017.

SEMESTER-VI

COURSE 5: QUANTUM ALGORITHMS

Theory

Credits: 4

5 hrs/week

Course Outcomes:

At the end of the course, students will be able to

CO1: Understand the fundamental differences between classical and quantum systems, including their computational and correlation properties.

CO2: Analyze the structure and behavior of quantum logic gates and circuits using geometric and matrix representations.

CO3: Apply universal gate sets to design and simulate basic quantum circuits.

CO4: Understand core principles and analyze quantum algorithm performance in comparison to classical counterparts across problem domains and

CO5: Study complexity and error correction

Unit 1: Qubits versus classical bits and Quantum correlations

Spin-half systems and photon polarizations, trapped atoms and ions, Artificial atoms using circuits, Semiconducting quantum dots, Single and Two qubit gates – Solovay - Kitaev Theorem, Entanglement and Bell's theorems, Time and space complexity (P, NP, PSPACE)

Unit 2: Universal quantum logic gates and circuits

Fundamentals of Quantum Gates: Definition and properties of quantum bits (qubits), Singlequbit gates (e.g., Pauli-X, Y, Z, Hadamard, Phase, T gates), Geometric representation of single qubit gates (Bloch Sphere), Multi-qubit gates (e.g., CNOT, CZ, Toffoli, Fredkin gates), Truth tables and matrix representations of common quantum gate, Universality of Quantum Gates: Definition of a universal set of quantum gates, Proof of universality for common sets (e.g., {H, T, CNOT}, {H, Phase, CNOT}, {Toffoli, Hadamard}), Solovay-Kitaev theorem and its implications for gate approximation, Approximating arbitrary unitary operations,

Unit 3. Quantum Circuit Design and Construction

Graphical representation of quantum circuits, Circuit decomposition and synthesis, Techniques for constructing multi-qubit gates from universal sets, Quantum Fourier Transform (QFT) circuit, Quantum phase estimation circuit, Shor's algorithm circuit components, Grover's algorithm circuit components

Unit 4: Quantum algorithms

Introduction to Quantum algorithms, Deutsch algorithm, Deutsch Josza algorithm, Bernstein - Vazirani algorithm, Simon's algorithm, Database search: Grover's algorithm, Quantum Fourier Transform and prime factorization: Shor's Algorithm. Advancements in Quantum Algorithms: Variational Quantum Eigensolver (VQE), HHL QAOA.

Unit 5: Quantum complexity and Error correction

Introduction to Quantum complexity and Quantum complexity classes: Q, EQP, BQP, BPP, QMA, Introduction to Error correction: Fault-tolerance, Simple error correcting codes, Survey of current status: NISQ era processors, Quantum advantage claims, Roadmap for future

Textbooks

1. Manenti R., Motta M., Quantum Information Science, 1st Edition, Oxford University Press, 2023.
2. Michael A. Nielsen, Isaac L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary Edition, Cambridge University Press, 2011.
3. Leonard Susskind, Art Friedman, Quantum Mechanics: The Theoretical Minimum, 1st Edition, Basic Books, 2014.
4. Jonathan A. Jones, Quantum Information, Computation and Communication, 1st Edition, Oxford University Press, 2012.
5. Phillip Kaye, Raymond Laflamme, Michele Mosca, An Introduction to Quantum Computing, 1st Edition, Oxford University Press, 2007.
6. Carl J. Pratt, Quantum Physics for Beginners, 2nd Edition, Independently Published, 2020.
7. Hui Yung Wong, Introduction to Quantum Computing: From a Layperson to a Programmer in 30 Steps, 1st Edition, Springer Nature Switzerland AG, 2022.

Reference Books

1. Eric R. Johnston, Nic Harrigan, Mercedes Gimeno-Segovia, Programming Quantum Computers, 1st Edition, O'Reilly Media, 2019.
2. Jack D. Hidary, Quantum Computing: An Applied Approach, 2nd Edition, Springer, 2021.
3. Mark M. Wilde, Quantum Information Theory, 2nd Edition, Cambridge University Press, 2017.
4. Eleanor Rieffel, Wolfgang Polak, Quantum Computation, 1st Edition, MIT Press, 2011.
5. A. Pathak, Elements of Quantum Computation and Quantum Communication, 1st Edition, CRC Press, 2015.
6. Quantum Computing Explained, David McMahon, 1st Edition, Wiley, 2008
7. Quantum Error Correction and Fault Tolerant Quantum Computing, Frank Gaitan, 1st Edition, CRC Press, 2008

SEMESTER-VI

COURSE 6: IBM QUANTUM COMPOSER AND QUANTUM COMPUTING USING QISKIT LAB

Theory

Credits: 4

5 hrs/week

Course Outcomes

CO1: Understand Quantum Computing Fundamentals

CO2: Design Quantum Gates

CO3: Develop Practical Skills with Quantum Tools

CO4: Apply Quantum Principles in Algorithms and Applications

CO5: Develop problem-solving skills essential for research in quantum machine learning, cryptography, and hybrid quantum-classical computing.

1. Explore the Functions of IBM Quantum Composer and Qiskit, PennyLane and Converse & QSim

Learn IBM's quantum composer in detail, it is a cloud-based visual interface that enables learners to design and simulate quantum circuits without requiring anyone to learn how to program. Moreover, it allows for a drag-and-drop interface that allows participants to learn about qubits, gates and measurements as intuitively as possible. Qiskit is an open-source quantum computing framework from IBM that gives programmers access to quantum processors as it is based on the Python programming language. The Indian QSim is a project by C-DAC, IISc, and IIT-Roorkee, designed to provide a platform for researchers and students to learn and experiment with quantum computing algorithms and hardware. PennyLane is a flexible and powerful open-source framework for quantum programming

- a) Understand the interface and features of IBM Quantum Composer.
- b) Write and execute simple quantum programs using Qiskit.
- c) Explore features of PennyLane and QSim

2. What are the Different Types of Quantum States?

Learn the core idea of quantum states, which, in a general sense, describe the state of a quantum system.

- a) Learn about pure states and what they mean for well-defined qubit states.
- b) Investigate superposition states, where the qubit or qubits exist in linear combinations of $|0\rangle$ and $|1\rangle$ at the same time, yielding the quantum computing potential of quantum parallelism.
- c) Learn about the entangled states, where the state of one qubit is combined with another qubit at any distance from the first qubit.
- d) Explore mixed states, represented by density matrices, which occasionally occur when a system is a probabilistic mixture of states due to noise and/or partial information about the quantum system.
- e) Differentiate between pure, superposition, entangled, and mixed states.

3. Single Qubit Gates

Single-qubit quantum gates, are the ones which map states of a single qubit such as $|0\rangle$ and $|1\rangle$ to other states. This is called the fundamental transformation on a single qubit. The foundational arithmetic-based Pauli gates (X, Y and Z) rotate a qubit about the x-y-z axes of the Bloch sphere, the Hadamard gate (H) creates a superposition of states $(|0\rangle + |1\rangle)/\sqrt{2} = |+\rangle$ or $(|0\rangle - |1\rangle)/\sqrt{2} = |-\rangle$, and the Phase (S or T) gates alter the relative phase on two states. Alone or in combination, the single-qubit gates build the foundation for more complex versions of quantum circuits and provide resolution for controlling quantum information.

- a) Apply single-qubit gates to transform basis states.
- b) Visualize gate operations on the Bloch sphere.
- c) Design simple quantum circuits using combinations of single-qubit gates.
- d) Demonstrate the role of Hadamard and Pauli gates in superposition and measurement.

4. Multi-Qubit Gates

Look into the class of gates that can perform interactions of either two or more qubits, which are necessary in order to utilize quantum algorithms and induce entanglement. Look at the CNOT (Controlled-NOT) gate, which flips a target qubit based on the state of a control qubit, and investigate the action of the SWAP gate, which exchanges the states of two qubits. Multi-qubit gates are necessary in order to be able to create entangled states, implement logic functions, and provide universal quantum computation.

- a) Implement two-qubit gates such as CNOT and SWAP.
- b) Demonstrate how multi-qubit gates generate entanglement.
- c) Use multi-qubit operations to build logic functions in quantum circuits.
- d) Recognize the role of entangling gates in quantum computation.

5. Control Gates

Investigate control gates which enable conditional execution of transformations. Control gates of single-qubit gates (e.g., Controlled-H, Controlled-Z) apply an operator to the target qubit if the control qubit is in a specific state. Study control gates which can be used to help understand the structure of quantum error correction codes, Grover's search algorithm, and Shor's factorization algorithm, and thus an essential phase in scalable and fault-tolerant quantum computation.

- a) Understand the function of controlled gates in quantum computation
- b) Construct circuits using controlled operations for conditional logic.
- c) Explain the importance of control gates in error correction and key algorithms.
- d) Model Grover's algorithm and Shor's algorithm building blocks with control gates.

6. Phase Gates and U Gates

Explore Phase gates, which introduce relative phase shifts in quantum states without affecting probability amplitudes, and study S and T gates as special cases of phase shifts. After phase gates, study U gates, which represent the most general operation that can be applied to a single qubit based on specifying three rotation parameters. The U gates extend the gate set of quantum gates to the more general unitary transformations, which are important for designing quantum algorithms and implementing quantum algorithms on a device-specific basis.

- a) Apply phase shifts to qubit states.
- b) Implement S, T, and U gates in Qiskit.

- c) Describe how U gates generalize all single-qubit operations.
- d) Connect phase gates to interference patterns in quantum algorithms.

7. Born Rule and Quantum Registers

Understand the value of the Born rule, which provides the formal mathematics for interpreting physical measurements in quantum trajectories as probabilities. Explore Quantum registers.

- a) Interpret measurement results using the Born rule.
- b) Construct quantum registers for storing multiple qubits.
- c) Analyze how measurement affects quantum information.
- d) Simulate probabilistic outcomes in multi-qubit systems.

8. Quantum Entanglement

Quantum entanglement occurs when the quantum state of one particle cannot be described independently of the state of another particle.

- a) Generate entangled states using Hadamard and CNOT gates.
- b) Explore teleportation
- c) Learn QKD protocols.

9. Unitary Representations and Bell States

Unitary operators play an important role in defining quantum gates and quantum evolutions. Bell states are a set of two-qubit states that are maximally entangled, with regard to defined sets of quantum entangled states values. Bell states are critical resources for quantum communication, no-locality experiments and many other applications. A detailed study of Bell states helps in the contexts of quantum cryptography, teleportation and many more.

- a) Represent quantum gates as unitary matrices.
- b) Derive and construct Bell states

10. Rotation Gates and Barrier Functions

Rotation gates (R_x , R_y , R_z) are important unitary gates because they give us fine-grain control over qubits. Specifically, they are very helpful because they allow us to control our qubits with rotations along axes of the Bloch sphere. They allow for continuous transformations and unlock rotations smaller than π . This is useful when we want to fine tune in QML.

Barrier functions indicate program control flow (as per your ordering in circuit layout) so operations have the ability to enforce both operation ordering and operation intersection, and stop compilers from switching operations or optimizing them out.

When directly coupled with rotation gates, barrier functions allow for both fewer errors and higher flexibility in the implementation of quantum circuits.

- a) Apply rotation gates to achieve precise qubit control.
- b) Implement variational circuits using rotation gates.
- c) Use barrier functions to enforce operation order in quantum programs.

References:

1. Qiskit: <https://qiskit.org/textbook/preface.html>
2. PennyLane: <https://pennylane.ai/>
3. CDAC QSim: https://www.cdac.in/index.aspx?id=product_details&productId=QSim